Regd. No. | | | | | | | | | | |

VIGNAN'S
FOUNDATION FOR SCIENCE, TECHNOLOGY & RESEARCH
(Deemed to be University) - Estd. u/s 3 of UGC Act 1956

Regulation: R22C24 **Course Code: 24PY201-2**

II B.Tech I Semester Supplementary Examinations – July, 2026

PHYSICS FOR QUANTUM COMPUTING

(CSE(IOT))

Time: 150 Min **Max. Marks:60M**

SECTION – A

Answer all Four questions **4X8M=32M**

1.

a) Explain polarization and types of polarization? [2M]

b) Describe the quantum superposition principle? [2M]

c) Explain the working principle of the three-qubit bit-flip error correction code? [4M]

2. For a given quantum state $|\psi_1\rangle = \alpha|00\rangle+\beta|10\rangle$ the probability amplitude satisfies, i.e., $|\alpha|^2 +|\beta|^2 = 1$.

a) Obtain the output after CNOT gate action on $|\psi_1\rangle$. [2M]

b) What are the final probabilities of the state resulting from CNOT gate action on $|\psi_1\rangle$? [3M]

c) What would be the output if H-gate acts only on second qubits of $|\psi_1\rangle$. [3M]

3. Consider the following two quantum states: $|\psi\rangle = \begin{pmatrix} 4i \\ 5 \\ -3i \end{pmatrix}$ and $|\phi\rangle = \begin{pmatrix} 7 \\ 2i \\ -9i \end{pmatrix}$

a) Are $|\psi\rangle$ and $|\phi\rangle$ normalized? If not, normalize it. [2M]

b) Are $|\psi\rangle$ and $|\phi\rangle$ orthogonal? [2M]

c) Calculate the inner and outer product of $|\psi\rangle$ and $|\phi\rangle$. [4M]

4.

a) What would be the output when CNOT gate followed by H $\otimes$ I acted on below quantum state $|\psi_2\rangle$ [3M]

$$|\psi_2\rangle = \frac{1}{\sqrt{3}}|00\rangle + \frac{\sqrt{2}}{\sqrt{3}}11\rangle$$

Here 'I' is the 1-qubit identity operation, H is the 1-qubit Hadamard, and CNOT is the 2-qubit controlled-not operation with the first qubit being the control.

b) What is the chance of $|\psi_2\rangle$ resulting in state $|11\rangle$ after the measurement? [2M]

c) Show that the commutator of two Hermitian operators is anti-Hermitian? [3M]

SECTION - B

Answer all Two questions **2X14M=28M**

5.

a) State Deutsch-Josza algorithm and prove it for quantum circuit for number of qubits two (n = 3). [8M]

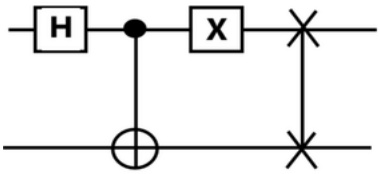
b) Compare and highlight the important differences of the classical bits and quantum bits (qubits) in the context of quantum computation. [2M]

c) Compare classical and quantum theory of light-matter Interaction and highlight its relevance to quantum computing. [4M]

6.

a) Estimate the output of given quantum circuits. [10M]

(i)



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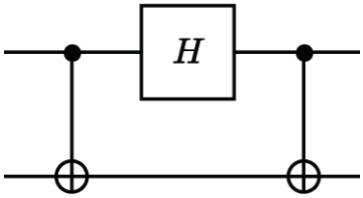

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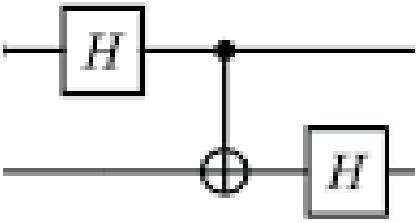

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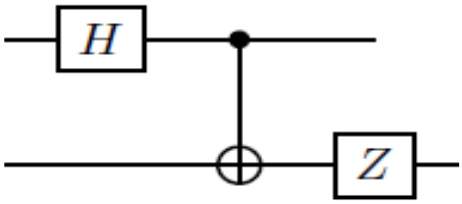

b) Elaborate Di Vincenzo criteria for realizing qubits. [4M]